

Middle Technical University

Electrical Engineering Technical College

Cyber Security Technology Engineering Department



COMMUNICATION FUNDAMENTALS

Second Stage

Basic Principles of Communication

Prepared by: Assist. Lec. Mustafa Ahmed Saadi

2025 – 2024

INTRODUCTION

Communication plays a crucial role in our daily lives and modern technology. By the end of this lecture, students will have a solid understanding of fundamental communication concepts. They will learn to describe the key components of a basic communication system and appreciate the significance of communication in telecommunication and cybersecurity. Understanding these principles is vital for designing, securing, and maintaining reliable communication networks.

What is Communication?

Communication is the process of exchanging information, ideas, or data between two or more entities using a defined medium. It is fundamental to human interaction and technological advancements. Communication can be verbal, written, visual, or electronic, and it is the backbone of many fields, including telecommunications, networking, and cybersecurity. Effective communication ensures that information is transmitted accurately, minimizing misunderstandings and errors.

Key Elements of Communication:

Communication systems rely on several essential components that ensure the successful transfer of information. These include:

1. **Sender:** The sender is the source of information. It could be a person, device, or system generating the message. For instance, in telecommunication, a sender could be a mobile phone transmitting a voice signal.
2. **Message:** The message is the data or information that needs to be transmitted. It can be text, audio, video, or other digital formats, depending on the communication system.

3. **Medium:** The medium is the channel through which the message travels. This can be physical, such as fiber optic cables or copper wires, or wireless, such as radio waves or infrared signals.
4. **Receiver:** The receiver is the entity that receives and interprets the message. In a telecommunication system, this could be a phone, computer, or television.
5. **Feedback:** In some communication systems, feedback is necessary to confirm that the message was received correctly. Feedback is crucial in interactive communication, such as video calls or messaging systems.
6. **Protocols:** Protocols are predefined rules and conventions that govern data exchange. They ensure proper synchronization between the sender and receiver, define error detection and correction mechanisms, and standardize data handling. Examples include Internet Protocol (IP), Transmission Control Protocol (TCP), Hypertext Transfer Protocol (HTTP), and Simple Mail Transfer Protocol (SMTP). These protocols facilitate interoperability between different systems and devices, enhancing reliability and security in communication.

Importance in Telecommunication:

Communication is a fundamental pillar of modern society, enabling essential services like the internet, mobile communication, and satellite systems. In cybersecurity, understanding communication systems is crucial for securing data and preventing unauthorized access. Cyber threats such as eavesdropping, interference, and hacking target communication networks, making it imperative to implement secure communication practices.

The Block Diagram of a Communication System:

A communication system consists of multiple interconnected components, each with a specific function. These components work together to ensure efficient and reliable information transmission. The following sections describe the primary components of a basic communication system.

Components of a Basic Communication System:

1. **Information Source:** The information source is where the message originates. It could be a human voice, a computer sending data, a sensor collecting measurements, or a video stream. The message is then prepared for transmission to the receiver.
2. **Transmitter:** The transmitter processes the message to make it suitable for transmission. This involves several functions:
 - **Modulation:** Modulation is the process of superimposing the message signal onto a carrier wave to facilitate efficient transmission over long distances.
 - **Encoding:** Encoding helps add error detection or correction codes to enhance data reliability.
 - **Amplification:** Amplification boosts the signal strength to ensure that it can travel long distances without significant degradation. Examples of transmitters include microphones, modulators, and amplifiers in communication devices.
3. **Transmission Medium:** The transmission medium is the channel that carries the signal from the transmitter to the receiver. It can be wired, such as fiber optic cables and coaxial cables, or wireless, such as radio waves and infrared signals. Important characteristics of transmission media include:

- **Bandwidth:** Determines the amount of data that can be transmitted over the medium.
 - **Noise Susceptibility:** Different media have varying levels of resistance to noise and interference, which can affect signal quality.
4. **Noise Source:** Noise is any unwanted signal that interferes with the transmission and reception of the message. Various types of noise can degrade communication quality:
- **Thermal Noise:** Random electronic activity within transmission components that introduces minor distortions.
 - **Interference:** Signals from other devices or systems that disrupt communication.
 - **Impulse Noise:** Sudden, short-duration noise caused by external factors like electrical surges or switching devices.
 - **Effects of Noise:** Noise can reduce the clarity and accuracy of the transmitted message, requiring noise reduction techniques such as filtering, shielding, or digital error correction.
5. **Receiver:** The receiver extracts and reconstructs the original message from the transmitted signal. This involves:
- **Demodulation:** Separating the message signal from the carrier wave.
 - **Decoding:** Converting encoded data back into its original form.
 - **Filtering:** Removing noise or interference to improve signal quality.
- Examples of receivers include antennas, demodulators, and audio or video output devices.

6. **Destination:** The destination is the final point where the information is delivered and interpreted. Examples include speakers that output sound, screens that display video, or computers that process received data.

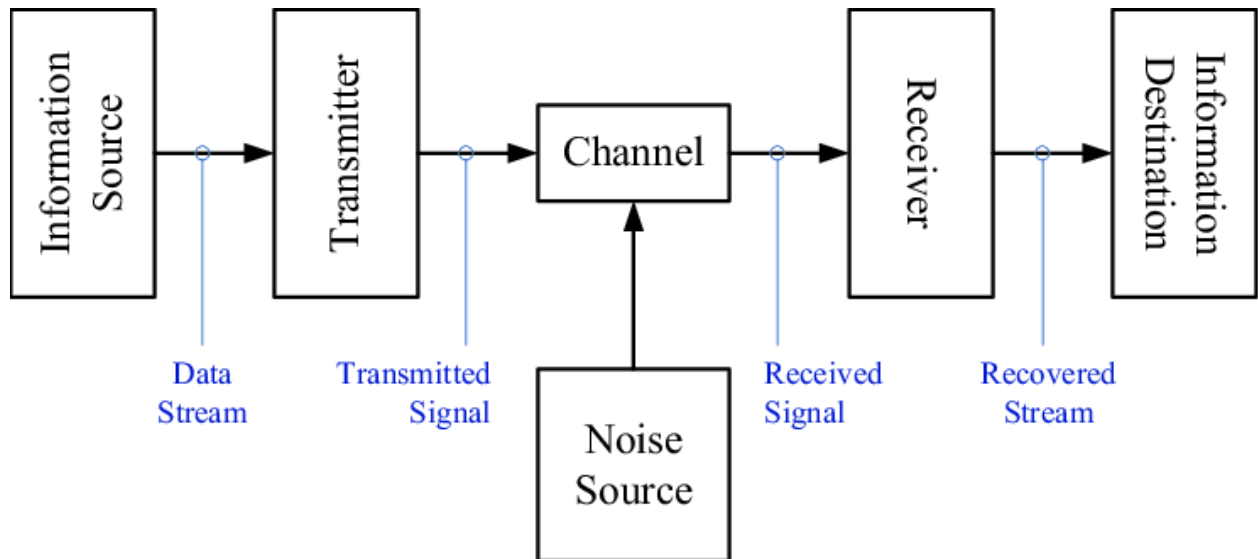


Figure 1: The Block Diagram of a Communication System

Types of Transmission Medium

1. Wired Communication:

Wired communication involves physical connections between transmitting and receiving devices. Examples include:

- **Telephony:** Converts voice into electrical signals for transmission through copper wires.
- **Fiber Optics:** Uses light signals to transmit high-speed data over long distances with minimal interference.

2. Wireless Communication:

Wireless communication systems use electromagnetic waves to transmit information. Examples include:

- **Mobile Phones:** Use radio waves for voice and data communication.
- **Satellite Communication:** Relays signals between Earth stations and satellites, enabling global connectivity.

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COMMUNICATION FUNDAMENTALS

Second Stage

Signals: Principles and Definition

Prepared by: Assist. Lec. Mustafa Ahmed Saadi

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Introduction

One of the most fundamental concepts in telecommunications is signal. Signals play a crucial role in the transmission of data and information across different systems, making them an essential topic for engineers working in communication networks and data transmission.

1. What Are Signals?

A signal is a function that conveys information regarding a physical phenomenon. It acts as a carrier of information, allowing communication between two or more points. Signals are an integral part of telecommunications systems and exist in various forms, including electrical signals, optical signals, and acoustic signals.

- **Electrical signals** refer to variations in voltage or current that represent information and are commonly used in telecommunication networks and electronic circuits.
- **Optical signals** use light waves to transmit information, such as in fiber optic communication systems where data is sent through pulses of light.
- **Acoustic signals** involve sound waves and are used in applications like sonar and audio communication.

In modern telecommunication systems, signals serve as the foundation for transmitting voice, video, and data across different media, ensuring seamless connectivity and efficient communication.

2. Principles of Signals

Understanding the fundamental principles of signals is crucial for designing and optimizing communication systems. Signals can be analyzed based on how they vary with time and frequency, as well as their inherent properties.

2.1 Representation of Signals

Signals can be represented in two primary domains:

- **Time domain representation** shows how a signal changes over time. This form of representation is useful for analyzing real-time behavior, waveforms, and the effects of noise on signals.
- **Frequency domain representation** examines a signal based on its frequency components. By using mathematical tools like the Fourier Transform, a signal can be broken down into its constituent frequencies, which is particularly useful in applications such as filtering and modulation.

2.2 Signal Properties

Several key properties define a signal and its behavior in a communication system:

- **Amplitude:** represents the strength or intensity of the signal. It determines the power level of the transmitted signal and affects how far it can travel without significant degradation.
- **Frequency:** is the number of oscillations or cycles a signal completes per second, measured in Hertz (Hz). Different frequencies are used for various communication applications, such as radio, television, and mobile networks.
- **Phase:** describes the position of a signal wave relative to a reference point in time. It plays a critical role in modulation techniques and synchronization in communication systems.

- **Bandwidth:** refers to the range of frequencies occupied by a signal. Higher bandwidth allows for faster data transmission, which is crucial in modern high-speed communication networks.

3. Types of Signals

Signals are generally classified into two main categories: analog signals and digital signals. Each type has distinct characteristics, advantages, and applications.

3.1 Analog Signals

Analog signals are continuous in nature and smoothly vary over time. They can take on an infinite number of values within a given range. Examples of analog signals include sound waves, temperature fluctuations, and electrical signals in traditional telephony.

Key characteristics of analog signals include:

- They have infinite resolution, meaning they can represent subtle variations in a phenomenon.
- Analog signals are highly susceptible to noise and distortion, which can degrade signal quality over long distances.
- Processing and storage of analog signals require specialized hardware and are more complex compared to digital signals.

3.2 Digital Signals

Digital signals, in contrast, are discrete in nature and are represented in binary form using 0s and 1s. These signals are widely used in modern computing, networking, and digital communication systems.

Key characteristics of digital signals include:

- They are less affected by noise and interference, making them more reliable for long-distance communication.
- Digital signals can be easily stored, processed, and encrypted, which is vital for cybersecurity applications.
- They have limited resolution since they represent information in discrete steps, but they can be enhanced through techniques like quantization and error correction.

4. Difference Between Analog and Digital Signals

To better understand the contrast between analog and digital signals, the following table highlights their key differences:

Aspect	Analog Signals	Digital Signals
Nature	Continuous waveforms	Discrete pulses
Representation	Infinite number of values	Finite (binary) values
Noise Sensitivity	Highly affected by noise	Less affected by noise
Bandwidth Usage	Typically requires more bandwidth	Efficient bandwidth usage
Processing	Difficult to process with computers	Easy to process with computers

Aspect	Analog Signals	Digital Signals
Storage	Difficult to store	Easy to store digitally
Example	AM/FM radio signals, thermometers	Ethernet data, MP3 files

Analog Signal Applications

Analog signals are still used in several important applications, including:

- Traditional broadcasting systems such as AM/FM radio and television.
- Analog telephone networks, where voice signals are transmitted as electrical waveforms.
- Biomedical devices like electrocardiograms (ECGs), which monitor heart activity in hospitals.

Digital Signal Applications

With the advancement of digital technology, digital signals have become the standard for most modern applications, including:

- Mobile networks and fiber-optic communication systems, which rely on digital signals for efficient data transmission.
- Computer networking protocols, such as TCP/IP, which enable reliable communication over the internet.
- Cryptographic techniques and secure communication, where digital signals are used to encrypt and protect sensitive information from cyber threats.

Digital signals are more resilient due to advanced error detection and correction techniques, making them preferable in high-security and high-reliability applications.

SIGNAL PROPERTIES

In analog signal processing, signals can be classified based on various properties such as periodicity, causality, power or energy, and determinism.

1. Periodicity

A signal is **periodic** if it repeats itself after a fixed time period T .

Mathematically, a signal $x(t)$ is periodic if:

$$x(t + T) = x(t) \quad \text{for all } t$$

Example of a periodic signal:

- A sinusoidal signal $x(t) = A \cos(\omega t + \phi)$ with period $T = 2\pi/\omega$
- A square wave with a fixed frequency
- **Example of a non-periodic signal:**
 - An exponentially decaying function $x(t) = e^{-t}$
 - A unit step function $u(t)$

2. Causality

A signal is **causal** if it is zero for all negative time values (i.e., it exists only for $t \geq 0$).

$$x(t) = 0, \quad \text{for } t < 0$$

- **Example of a causal signal:**
 - Unit step function: $u(t) = 1, \quad \text{for } t \geq 0$ $u(t) = 0 \quad \text{for } t < 0$
 - Exponentially decaying function: $x(t) = e^{-t} u(t)$

- **Example of a non-causal signal:**
 - A sine wave: $x(t) = \sin(t)$, which exists for all t
 - A Gaussian function centered at $t = 0$

3. Power or Energy Signal

A signal is classified as either a **power signal** or an **energy signal** based on its energy and power calculations:

- **Energy of a signal** $x(t)$ is given by:

$$E = \int_{-\infty}^{\infty} |x(t)|^2 dt$$

If E is finite, the signal is an **energy signal**.

- **Power of a signal** is given by:

$$\lim_{T \rightarrow \infty} \frac{1}{2T} \int_{-T}^T |x(t)|^2 dt$$

If P is finite and nonzero, the signal is a **power signal**.

- **Examples:**
 - **Energy signal:** A decaying exponential $x(t) = e^{-t}u(t)$ (finite energy, zero power)
 - **Power signal:** A sinusoidal wave $x(t) = A \cos(\omega t)$ (infinite energy, finite power)

4. Deterministic or Non-Deterministic (Random) Signal

- A **deterministic signal** can be exactly described by a mathematical function.
- A **non-deterministic (random) signal** cannot be predicted precisely and has a probabilistic nature.
- **Examples:**
 - **Deterministic signal:** A sine wave $x(t) = A\cos(t)$
 - **Non-deterministic signal:** A random noise signal in a communication system

5. Even and Odd Signals

- A signal is **even** if it is symmetric around the vertical axis:

$$x(t) = x(-t)$$

Example: $x(t) = \cos(t)$ (cosine function is even).

- A signal is **odd** if it has symmetry about the origin:

$$x(t) = -x(-t)$$

Example: $x(t) = \sin(t)$ (sine function is odd).

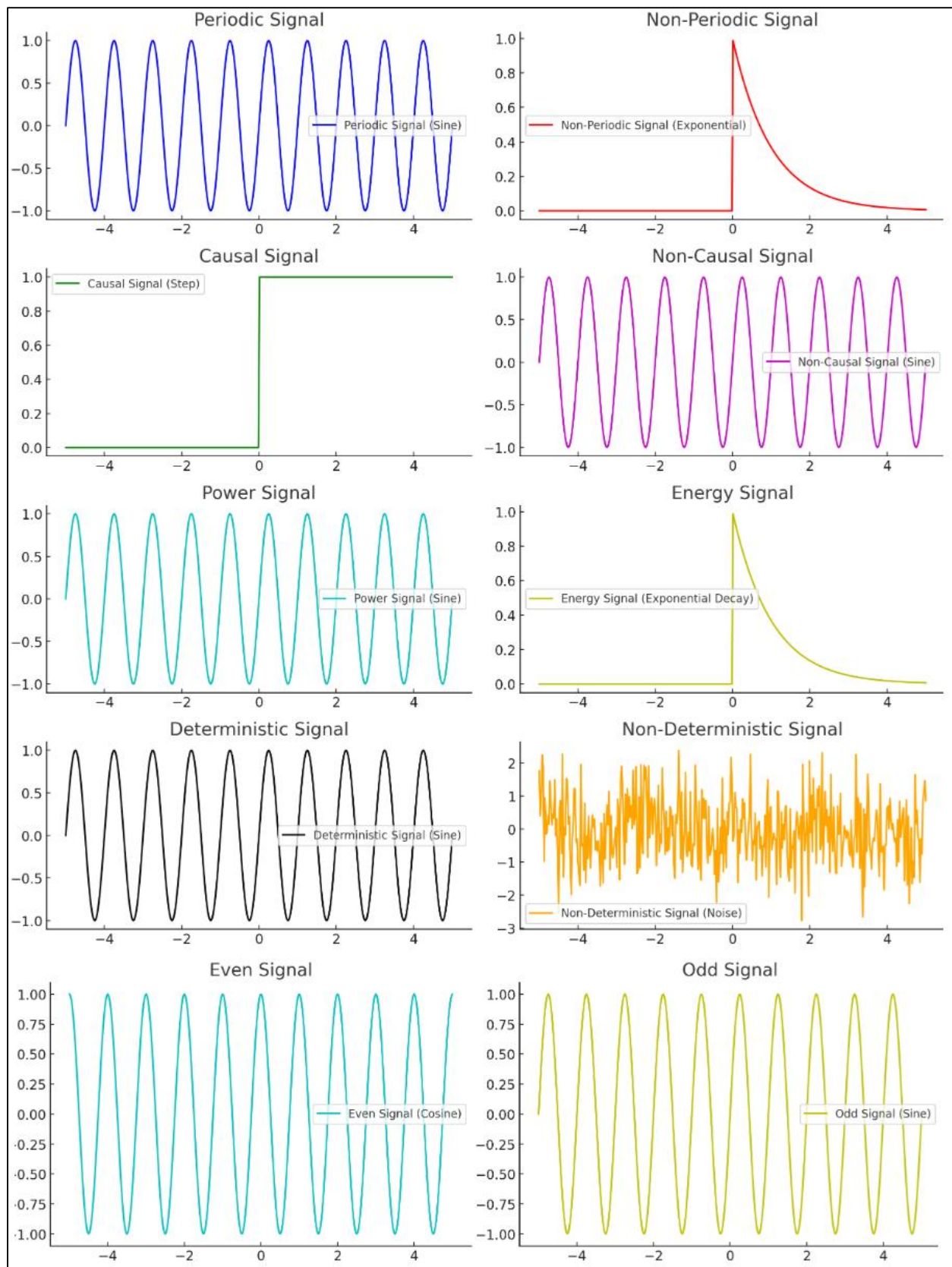


Figure 1: Graphical representation of the properties

Operations on signals

1. Amplitude-Scaling of Signal

Consider a signal $x(t)$ which is multiplying by a constant ' A ' and this can be indicated by a notation $x(t) \rightarrow Ax(t)$. For any arbitrary ' t ' this multiplies the signal value $x(t)$ by a constant ' A '. Thus, $x(t) \rightarrow Ax(t)$ multiplies $x(t)$ at every value of ' t ' by a constant ' A '. This is called amplitude-scaling. If the amplitude-scaling factor is negative then it flips the signal with the t -axis as the rotation axis of the flip. If the scaling factor is -1 then only the signal will be flip. This is shown in the Figure (a), (b), (c) which is given below.

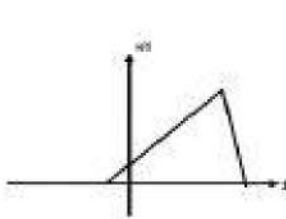


Fig.6(a) A signal $x(t)$

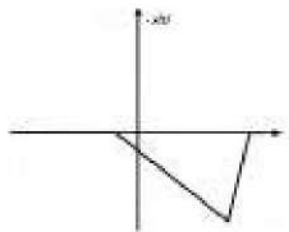


Fig.6(b) A signal $x(t)$ scaled by -1

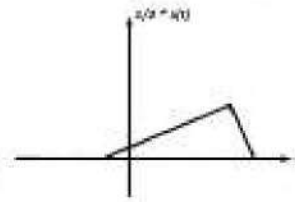


Fig.6(c) A signal $x(t)$ scaled by $1/2$

2. Time-Shifting of Signal

In signals and system amplitude scaling, time shifting and time scaling are some important properties. If a continuous time signal is defined as $x(t) = s(t - tI)$. Then we can say that $x(t)$ is the time shifted version of $s(t)$.

Consider a simple signal $s(t)$ for $0 < t < 1$

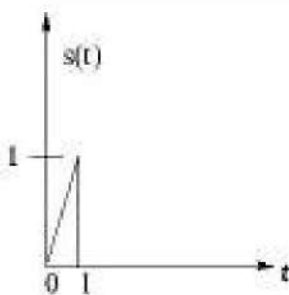


Fig.8(a) Signal within $0 < t < 1$

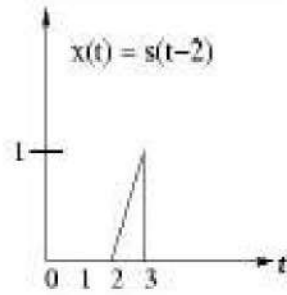


Fig.8(b) Signal shifted by 2 sec.

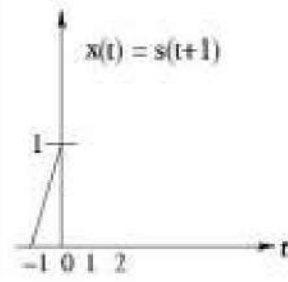


Fig.8(c) Signal shifted by -1 sec.

Now shifting the function by time $tI = 2$ sec.

$$x(t) = s(t-2) = t-2 \quad \text{for } 0 < (t-2) < 1 \\ = t-2 \quad \text{for } 2 < (t-2) < 3$$

Which is simply signal $s(t)$ with its origin delayed by 2 sec.

Now if we shift the signal by $tI = -1$ sec.

$$\text{then } x(t) = s(t+1) = t+1 \quad \text{for } 0 < (t+1) \\ = t+1 \quad \text{for } -1 < t < 0.$$

Which is simply $s(t)$ with its origin shifted to the left or advance in time by 1 seconds. This time-shifting property of signal is shown in the Figure 8(a), 8(b) and 8(c) given above.

3. Time-Scaling of Signal

Time scaling compresses or dilates a signal by multiplying the time variable by some quantity. If that quantity is greater than one, the signal becomes narrower and the operation is called **compression**. If that quantity is less than one, the signal becomes wider and the operation is called **dilation or (Expansion)**. Figure 7(a), 7(b), 7(c) shows the signal $x(t)$, compression of signal and dilation of signal respectively.

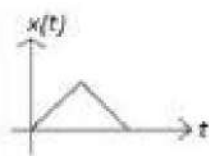


Fig.7(a) Signal $x(t)$

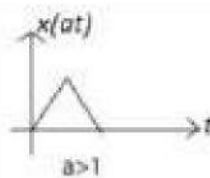


Fig.7(b) Compression of signal

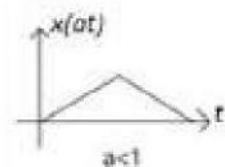
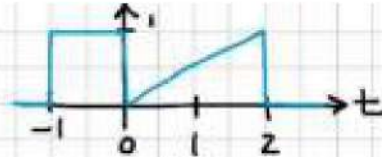
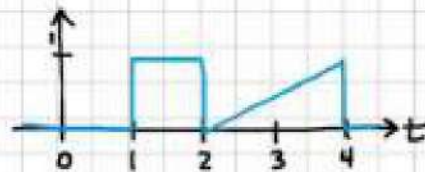


Fig.7(c) Dilation of signal

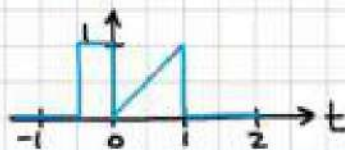
Example (1)

Consider the signal 
Sketch the following:

a) $x_1(t) = x(t-2) \Rightarrow$ Time shift 2s to the right
(i.e. delay)



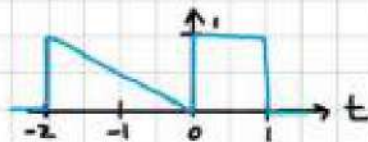
b) $x_2(t) = x(2t) \Rightarrow$ Compress in time by factor of 2.



c) $x_3(t) = x(0.2t) \Rightarrow$ Expand in time by factor of $\frac{1}{0.2} = 5$



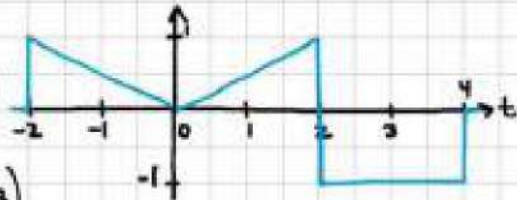
d) $x_4(t) = x(-t) \Rightarrow$ Reverse in time



Example (2)

Signal Operations #2

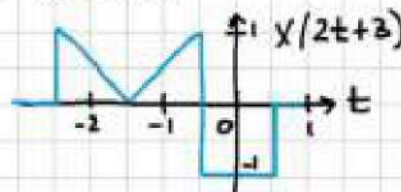
Consider the signal $x(t)$



Sketch the signal $y(t) = x(2t+3)$

Approach #1

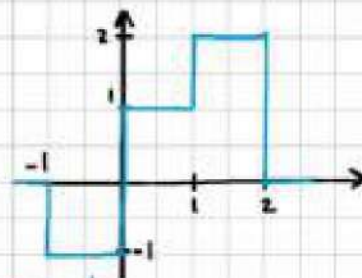
$$x(t) \xrightarrow{\text{Time shift}} x(t+3) \xrightarrow{\text{Time (compression) scale}} x(2t+3)$$



Example (3)

Signal Operations #3

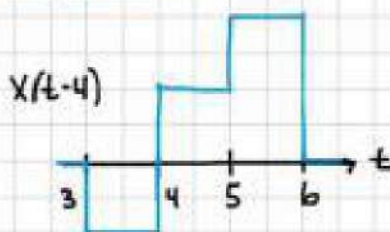
Consider the signal $x(t)$



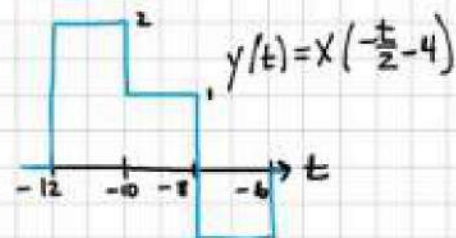
Sketch $y(t) = x(-\frac{t}{2}-4)$

Approach #1

$$x(t) \xrightarrow{\text{Time Shift}} x(t-4) \xrightarrow{\text{Flip and expand}} x(-\frac{t}{2}-4)$$



$$\begin{aligned} 6 &\rightarrow -12 \\ 5 &\rightarrow -10 \\ 4 &\rightarrow -8 \\ 3 &\rightarrow -6 \end{aligned}$$



COMMUNICATION FUNDAMENTALS

Second Stage

Modulation and Demodulation: needs and methods

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INTRODUCTION

- **Modulation** is the process of modifying a carrier signal according to the information signal (message) to facilitate efficient transmission over long distances.
- The primary purpose of modulation is to adapt the message signal for transmission over a medium by altering one or more properties of a high-frequency carrier signal. This ensures that the signal can travel long distances without significant degradation, allows multiple signals to share the same channel, and enhances resistance to noise and interference.

- Modulation is a fundamental technique in modern communication systems, playing a crucial role in radio, television, mobile networks, and satellite communication. Without modulation, low-frequency signals, such as human speech or data, would be ineffective for transmission as they suffer from attenuation and interference. By converting these signals to higher frequencies, modulation enables effective and reliable communication across various platforms and environments.
- On the receiving end, **demodulation** is the reverse process, where the original message signal is extracted from the modulated carrier. This ensures that the intended information is accurately retrieved and interpreted by the receiver, making modulation and demodulation an integral pair in communication technology.

Basic Concepts

- A **carrier signal** is a high-frequency waveform that carries the actual information. It is a continuous wave that is modulated in amplitude, frequency, or phase to encode the message signal.
- The **message signal**, also known as the baseband signal, is the original information that needs to be transmitted. It can be in the form of voice, video, or data.
- The **modulator** is a device or circuit that modifies the carrier signal according to the message signal. It performs amplitude, frequency, or phase modulation based on the type of modulation technique used.

- The **demodulator**, on the other hand, is a device or circuit that extracts the original message signal from the modulated carrier at the receiving end. It reverses the modulation process to retrieve the transmitted information.
- **Bandwidth** refers to the range of frequencies required for transmitting the modulated signal without significant loss of information. Different modulation techniques require different bandwidths.
- **Signal-to-noise** ratio (SNR) is a measure of the signal strength relative to background noise. A higher SNR indicates better transmission quality. Multiplexing is a technique that allows multiple signals to be transmitted over the same communication channel simultaneously, improving efficiency.

Role in Communication Systems

- Modulation is essential in communication systems for several reasons:
 1. It allows signals to be transmitted over long distances, reduces interference and noise, and enables multiple signals to coexist in the same medium.
 2. modulation facilitates multiplexing for efficient communication and improves security and data integrity.
 3. Without modulation, communication would be unreliable and inefficient, limiting the effectiveness of modern telecommunication networks.

Modulation Needs and Methods

Modulation is necessary for several reasons:

1. Low-frequency signals such as human voice cannot travel long distances without significant loss. Modulation shifts these signals to higher frequencies, making long-distance transmission feasible.
2. Modulation enables multiplexing, allowing different signals to be transmitted simultaneously without interference.
3. High-frequency transmission is less susceptible to external noise and distortions, ensuring better signal quality.
4. Modulation reduces the size of antennas, as the size of an antenna is inversely proportional to the frequency of transmission. Higher frequency signals allow for smaller and more practical antenna designs.

Types of Modulation

- Modulation techniques can be broadly classified into analog and digital modulation methods.
- In analog modulation, the properties of a continuous carrier wave are modified based on the input signal.
- **Amplitude Modulation (AM)** varies the amplitude of the carrier wave while keeping its frequency and phase constant. It is widely used in radio broadcasting but is highly susceptible to noise.
- **Frequency Modulation (FM)** varies the frequency of the carrier wave while maintaining constant amplitude. FM provides better noise immunity than AM and is commonly used in high-fidelity audio transmission.
- **Phase Modulation (PM)** alters the phase of the carrier wave in accordance with the message signal. It is closely related to FM and is frequently used in digital data transmission.

- Pulse modulation techniques modify the properties of discrete pulses rather than continuous waves.
- **Pulse Amplitude Modulation (PAM)** changes the amplitude of the pulses according to the message signal, making it useful in digital and analog applications.
- **Pulse Width Modulation (PWM)** varies the duration of pulses while keeping amplitude constant, commonly used in motor control and power electronics.
- **Pulse Position Modulation (PPM)** alters the timing of pulses, which enhances noise immunity and is widely used in optical and radar communication.
- **Pulse Code Modulation (PCM)** is a digital modulation technique where the message signal is converted into binary code before transmission. PCM provides high accuracy and is fundamental in digital telephony and modern communication systems.

Applications of Different Modulation Methods

- Amplitude Modulation (AM) is widely used in radio broadcasting, public address systems, and aviation communication due to its simplicity and ability to transmit long distances. However, AM is highly susceptible to noise and interference.
- Frequency Modulation (FM), on the other hand, is preferred for FM radio broadcasting, television sound transmission, and two-way communication systems because it offers better noise immunity and higher sound quality compared to AM.
- Phase Modulation (PM) is commonly used in digital data transmission, satellite communications, and mobile communication systems. It provides robustness against noise and allows for higher data transmission rates.

- Pulse Amplitude Modulation (PAM) finds applications in Ethernet communication, speech transmission, and LED lighting due to its simplicity and moderate bandwidth requirements.
- Pulse Width Modulation (PWM) is extensively used in motor speed control, LED dimming, and power regulation applications. It provides efficient control of power in various electronic and industrial applications.
- Pulse Position Modulation (PPM) is utilized in optical communication, remote control systems, and radar applications due to its high noise immunity and low bandwidth requirements.
- Lastly, Pulse Code Modulation (PCM) is a fundamental technique in digital telephony, audio recording, and video transmission systems. It offers high data integrity and is widely used in modern digital communication systems.

Modulation Type	Parameter Modified	Noise Immunity	Bandwidth Requirement	Power Efficiency	Complexity	Application
AM	Amplitude	Low	Low	Low	Simple	Radio, TV
FM	Frequency	High	High	Moderate	Moderate	Radio, Communication
PM	Phase	High	Moderate	High	Complex	Data Transmission
PAM	Pulse Amplitude	Moderate	Moderate	Moderate	Simple	Audio, Ethernet
PWM	Pulse Width	High	Moderate	High	Moderate	Power control, LED dimming
PPM	Pulse Position	High	Low	High	Complex	Radar, Optical communication
PCM	Digital Encoding	Very High	High	High	Most Complex	Voice, Data

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Amplitude Modulation

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AMPLITUDE MODULATION

Amplitude Modulation (AM) is a technique used in electronic communication, most commonly for transmitting information via radio waves. In AM, the amplitude (strength) of the carrier wave is varied in proportion to the amplitude of the message signal (such as audio or video signals), while the frequency and phase remain constant.

Key Features of AM:

1. **Carrier Wave:** A high-frequency signal that carries the information.
2. **Modulating Signal:** The actual information (such as voice or music) that modifies the carrier wave.
3. **Sidebands:** AM produces two sidebands—an upper sideband (USB) and a lower sideband (LSB), which contain the same information.
4. **Bandwidth:** The bandwidth of an AM signal is twice the frequency of the modulating signal.

Types of AM:

1. **Double Sideband (DSB-AM):** Both upper and lower sidebands are transmitted along with the carrier.
2. **Single Sideband (SSB-AM):** Only one sideband (USB or LSB) is transmitted to save bandwidth.
3. **Vestigial Sideband (VSB-AM):** A portion of one sideband is suppressed, often used in TV broadcasts.

Applications of AM:

- AM radio broadcasting (medium-wave and short-wave radio)

- Aviation communication
- Two-way radio systems
- TV video signal transmission (VSB-AM)

Frequency Range of Amplitude Modulation (AM)

Amplitude Modulation (AM) is widely used in broadcasting, aviation communication, and some military applications. The frequency range of AM transmission depends on the application:

1. Standard AM Radio Broadcasting:

- **Carrier Frequencies:** 530 kHz – 1700 kHz
- **Sideband Frequencies:** Extend above and below the carrier by the modulating frequency

2. Aircraft Communication (Amplitude Modulated VHF Band):

- **Carrier Frequencies:** 118 MHz – 137 MHz

3. Shortwave AM Radio (HF Band):

- **Carrier Frequencies:** 1.6 MHz – 30 MHz

Bandwidth of Amplitude Modulation (AM)

The bandwidth of an AM signal is determined by the highest frequency present in the modulating signal (f_m):

$$\text{Bandwidth(BW)} = 2 * f_m$$

Where:

f_m = Maximum frequency of the modulating signal

BW = Total bandwidth of the AM signal

Example 1:

If an audio signal has a frequency range of **300 Hz to 5 kHz** find its bandwidth

$$BW = 2 \times 5\text{kHz} = 10\text{kHz}$$

This is the typical bandwidth allocated for AM radio stations.

MATHEMATICAL BREAKDOWN OF AMPLITUDE MODULATION (AM)

In amplitude modulation, the instantaneous amplitude of the carrier signal is varied in proportion to the message signal. Let's define the key components mathematically:

1. Carrier Wave

A high-frequency sinusoidal signal:

$$c(t) = V_c \cos(2\pi f_c t)$$

where:

V_c = Amplitude of the carrier wave

f_c = Carrier frequency

t = Time

2. Modulating (Message) Signal

The information signal that contains the desired message:

$$m(t) = V_m \cos(2\pi f_m t)$$

where:

V_m = Amplitude of the message wave

f_m = message frequency

3. Amplitude Modulated (AM) Signal

The AM signal is formed by varying the carrier amplitude according to the message signal:

$$V_{AM}(t) = [V_c + m(t)]\cos(2\pi f_c t)$$

Substituting $m(t)$:

$$V_{AM}(t) = \underbrace{[V_c + V_m \cos(2\pi f_m t)]}_{\text{envelope}} \underbrace{\cos(2\pi f_c t)}_{\text{carrier}}$$

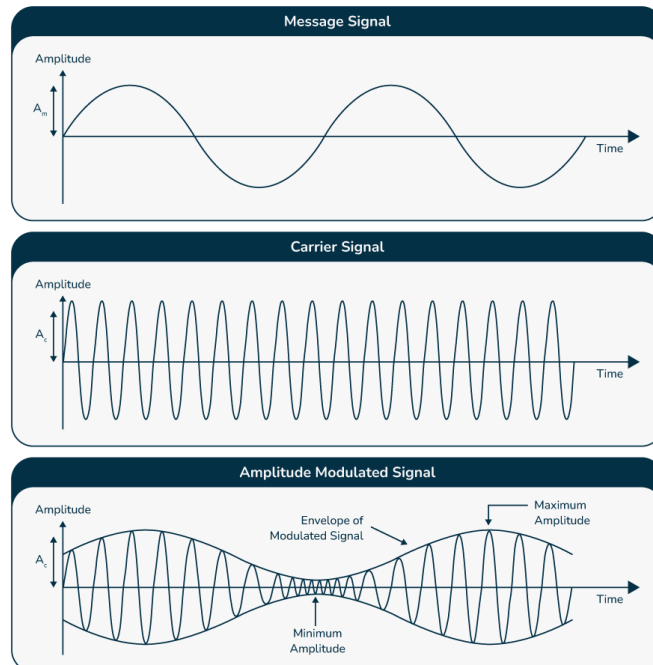
Expanding using trigonometric identities:

$$V_{AM}(t) = V_c \cos(2\pi f_c t) + V_m \cos(2\pi f_m t)\cos(2\pi f_c t)$$

Using the identity:

$$\cos(A)\cos(B) = \frac{1}{2} [\cos(A + B) + \cos(A - B)]$$

we get:



$$V_{AM}(t) = \underbrace{V_c \cos(2\pi f_c t)}_{\text{Carrier component}} + \underbrace{\frac{V_m}{2} \cos((f_c + f_m)2\pi t)}_{\text{Lower sideband component}} + \underbrace{\frac{V_m}{2} \cos((f_c - f_m)2\pi t)}_{\text{Upper sideband component}}$$

Frequency Spectrum of AM Signal

From the above equation, the AM signal consists of three frequency components:

- Carrier frequency(f_c) (with amplitude V_c)
- Upper sideband (USB):

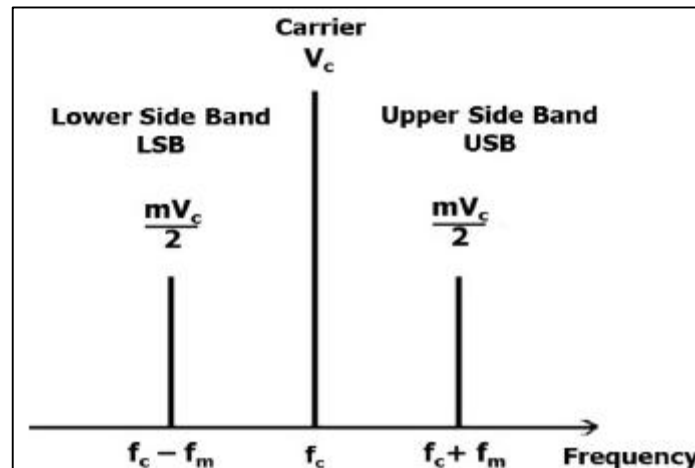
$$f_{USB} = f_c + f_m$$

$$V_{USB} = \frac{V_m}{2}$$

- Lower sideband (LSB):

$$f_{LSB} = f_c - f_m$$

$$V_{LSB} = \frac{V_m}{2}$$



Thus, the **total bandwidth** of the AM signal is:

$$\text{Bandwidth(BW)} = 2 * f_m$$

Which is **twice the highest frequency of the modulating signal**.

Modulation Index of Amplitude Modulation (m)

The **modulation index** in **Amplitude Modulation (AM)** is a measure of how much the carrier wave is varied by the modulating signal. It is denoted by **m** and is defined as:

$$m = \frac{V_m}{V_c}$$

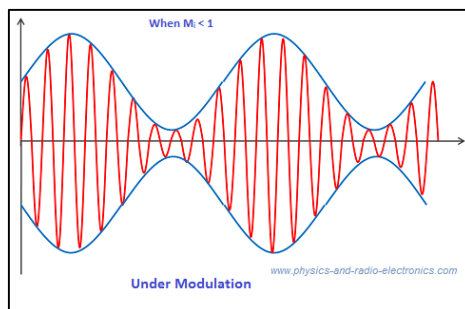
V_m = Peak amplitude of the modulating signal

V_c = Peak amplitude of the carrier signal

Interpretation of Modulation Index:

1. $m < 1$ (Under-modulation)

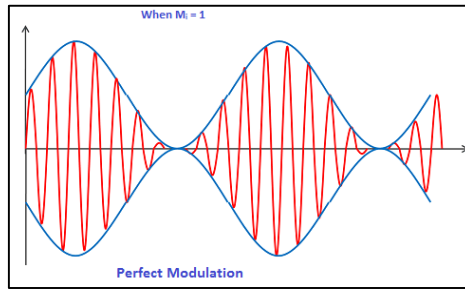
- The carrier is not fully utilized, and the signal may be weak.
- The message signal can be recovered easily.



2. $m = 1$ (100% Modulation, Critical Modulation)

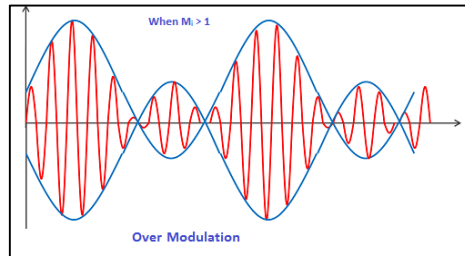
- The carrier amplitude varies from zero to twice its unmodulated value.

- This is the ideal case for efficient transmission without distortion.



3. $m > 1$ (Over-modulation)

- This causes **distortion** because the envelope of the AM signal is no longer a faithful reproduction of the modulating signal.
- Over-modulation leads to **sideband overlap**, causing **signal interference** and **loss of information** (also known as **envelope distortion**).



Alternative Expression Using Maximum and Minimum Amplitude

Sometimes m is calculated using the maximum ($V_{\max}$) and minimum ($V_{\min}$) values of the modulated signal envelope:

$$m = \frac{V_{\max} - V_{\min}}{V_{\max} + V_{\min}}$$

where:

- $V_{\max}$ = Maximum envelope amplitude

- $V_{\min}$ = Minimum envelope amplitude

Practical Considerations

- **Broadcast AM stations** typically use a modulation index around **0.8 to 1** to avoid distortion.
- **Over-modulation ($m > 1$)** should be avoided to prevent signal interference.

Amplitude Modulated (AM) Signal in terms of modulation signal:

As it mentioned before, the amplitude signal can be written as:

$$V_{AM}(t) = \underbrace{V_c \cos(2\pi f_c t)}_{\text{Carrier component}} + \underbrace{\frac{V_m}{2} \cos((f_c + f_m)2\pi t)}_{\text{Lower sideband component}} + \underbrace{\frac{V_m}{2} \cos((f_c - f_m)2\pi t)}_{\text{Upper sideband component}}$$

By multiplying the lower and upper sideband components by $\frac{V_c}{V_c}$

$$V_{AM}(t) = V_c \cos(2\pi f_c t) + \frac{V_m}{2} * \frac{V_c}{V_c} * \cos((f_c + f_m)2\pi t) + \frac{V_m}{2} * \frac{V_c}{V_c} * \cos((f_c - f_m)2\pi t)$$

$$\text{As } m = \frac{V_m}{V_c}$$

Then:

$$V_{AM}(t) = V_c \cos(2\pi f_c t) + \frac{V_c * m}{2} \cos((f_c + f_m)2\pi t) + \frac{V_c * m}{2} \cos((f_c - f_m)2\pi t)$$

Power in Amplitude Modulation

In Amplitude Modulation (AM), the transmitted signal consists of a carrier and two sidebands. As you now that the power is:

$$p = \frac{V^2}{R}$$

Then by dividing $V_{AM}(t)$ by R we will get the total power (P_{total}) of the amplitude modulated signal:

$$P_{total} = \frac{V_c^2}{R} + \frac{\left(\frac{V_c * m}{2}\right)^2}{R} + \frac{\left(\frac{V_c * m}{2}\right)^2}{R}$$

$$P_{total} = \frac{V_c^2}{R} + \frac{V_c^2 * m^2}{4R} + \frac{V_c^2 * m^2}{4R}$$

$$P_{total} = P_c + \frac{P_c * m^2}{4} + \frac{P_c * m^2}{4}$$

$$P_{total} = P_c \left(1 + \frac{m^2}{2}\right)$$

Thus, the total power in an AM signal is the sum of the carrier power and the power in the sidebands. Below are the equations of carrier power, Power of upper sideband, Power of lower sideband, and Power of sidebands.

$$P_c = \frac{V_c^2}{R} \dots\dots\dots \text{Carrier power}$$

$$P_{USB} = \frac{P_c * m^2}{4} \dots\dots\dots \text{Power of upper sideband}$$

$$P_{LSB} = \frac{P_c * m^2}{4} \dots\dots\dots \text{Power of lower sideband}$$

$$P_{SB} = P_{LSB} + P_{USB} = \frac{P_c * m^2}{2} \dots\dots\dots \text{Power of sidebands}$$

$$P_{total} = P_c + P_{SB} \dots\dots\dots \text{Total power}$$

Efficiency of Amplitude Modulation (AM)

The efficiency of AM transmission is the percentage of the total power that is carried by the sidebands (which contain the actual message information). It is defined as:

$$\eta = \frac{P_{SB}}{P_{total}} \times 100\%$$

$$\eta = \frac{\frac{P_c * m^2}{2}}{P_c \left(1 + \frac{m^2}{2}\right)} \times 100\%$$

$$\eta = \frac{m^2}{2 + m^2}$$

EXAMPLE 2:

An AM Transmitter of total power 100 watt, using percentage modulation 100%, find:

1. Power of carrier (P_c).
2. Power in side bands.

SOLUTION:

$$1. P_{total} = P_c \left(1 + \frac{m^2}{2}\right)$$

$$100 = P_c \left(1 + \frac{1}{2}\right)$$

$$P_c = 66.6 \text{ watt}$$

$$2. P_{total} = P_c + P_{SB}$$

$$100 = 66.6 + P_{SB},$$

$$P_{SB} = 33.4 \text{ watt}$$

EXAMPLE 3:

A broadcast radio transmitter radiates 10 watt, the modulation percentage is 60%, calculate the power carrier.

SOLUTION:

$$P_{total} = P_c \left(1 + \frac{m^2}{2} \right)$$

$$10 = P_c \left(1 + \frac{(0.6)^2}{2} \right)$$

$$P_c = 5.95 \text{ watt}$$

EXAMPLE 4:

A 1000 Hz carrier is simultaneously modulated with 300 Hz, 800 Hz radio sine wave what will be the frequencies present in the output?

SOLUTION:

A. When $f_m = 300 \text{ Hz}$

$$f_{LSB} = f_c - f_m$$

$$f_{LSB} = 1000 - 300 = 700 \text{ Hz}$$

$$f_{USB} = f_c + f_m$$

$$f_{USB} = 1000 + 300 = 1300 \text{ Hz}$$

B. When $f_m = 800 \text{ Hz}$

$$f_{LSB} = f_c - f_m$$

$$f_{LSB} = 1000 - 800 = 200 \text{ Hz}$$

$$f_{USB} = f_c + f_m$$

$$f_{USB} = 1000 + 800 = 1800 \text{ Hz}$$

EXAMPLE 5:

A spectrum analysis of an AM wave produces the following information:
50V at 750 KHz, 20V at 748 KHz, 20V at 752 KHz, find

- a. carrier frequency
- b. sideband frequencies
- c. modulating signal frequency
- d. modulation factor

SOLUTION:

a. $f_c = 750 \text{ KHz}$

b. $f_{LSB} = 748 \text{ KHz}$

$$f_{USB} = 752 \text{ KHz}$$

c. $f_{USB} = f_c + f_m$

$$752 = 750 + f_m$$

$$f_m = 2 \text{ KHz}$$

d. $m = \frac{V_m}{V_c}$

$$V_c = 50 \text{ volt}$$

$$V_{LSB} = V_{USB} = \frac{V_m}{2}, \quad 20 = \frac{V_m}{2}, \quad V_m = 40 \text{ volt}$$

$$m = \frac{40}{50} = 0.8$$



Amplitude Modulation – continuous

Assist. Lect. Mustafa Ahmed Saadi

OBJECTIVES

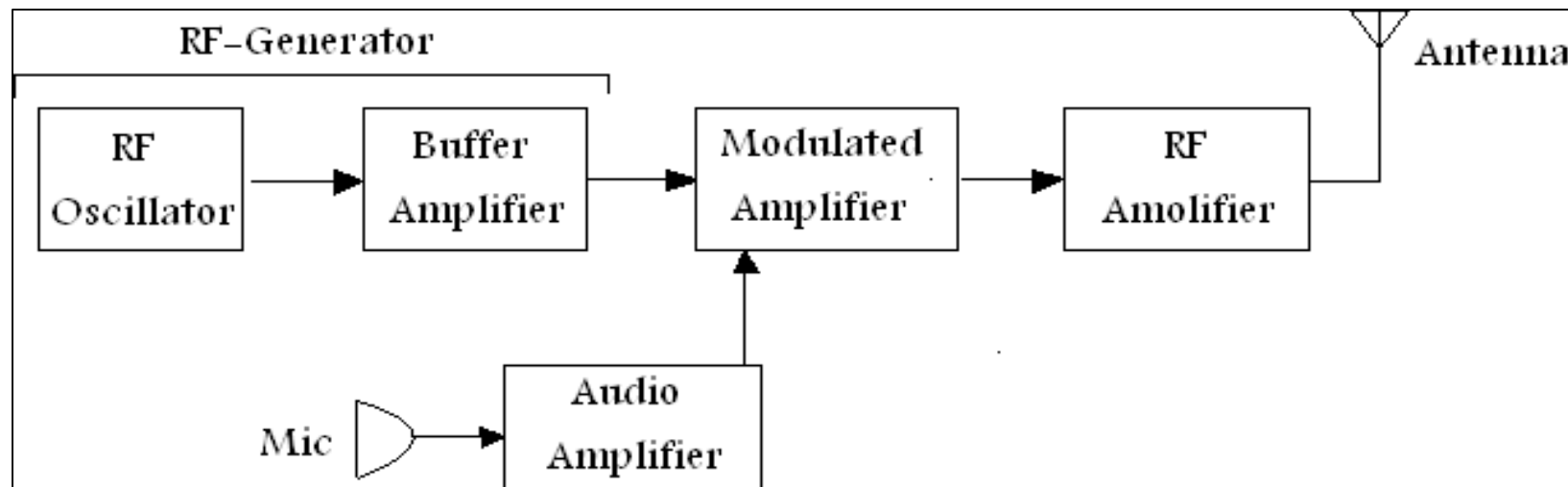
By the end of this lesson, students will be able to:

1. Understand the Transmitter Block Diagram.
2. Describe Generation techniques of Amplitude.
3. Understand Radio Receiver Essential Parameters of Amplitude Modulation.
4. Understand Demodulation process of AM Signal

Transmitter Block Diagram

RF oscillator: used to generate carrier signal.

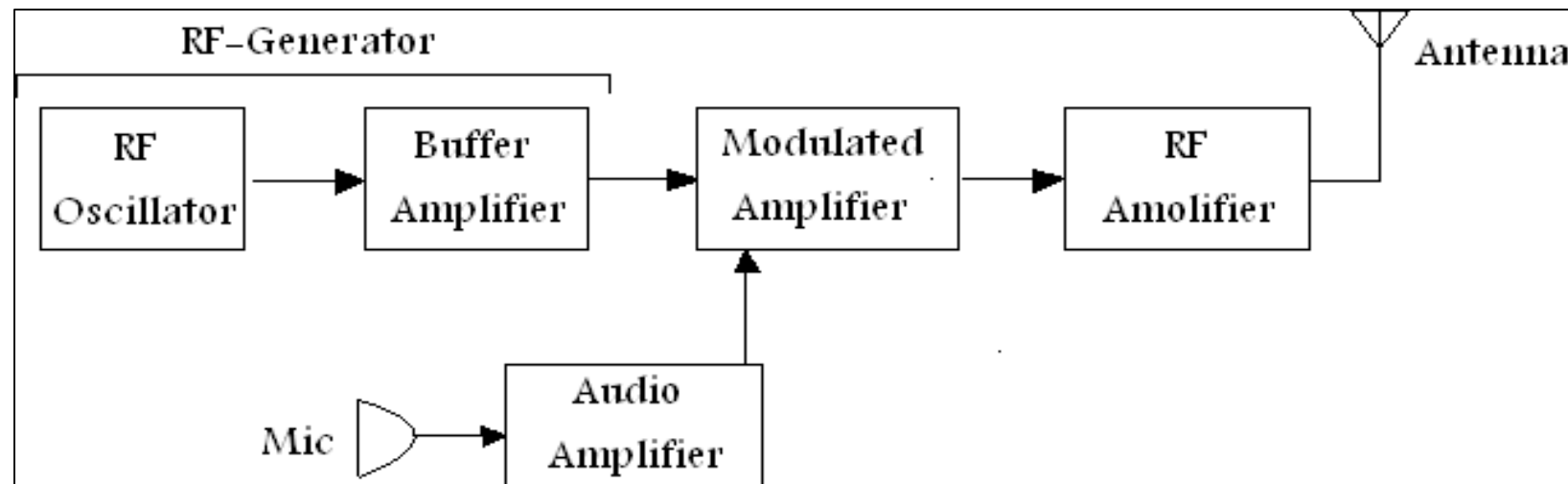
Buffer Amplifier: Used to isolate the RF oscillator from any interference from the rest of the Tx in order to keep a stable carrier frequency from the oscillator, so as to keep constant spectrum.



Transmitter Block Diagram

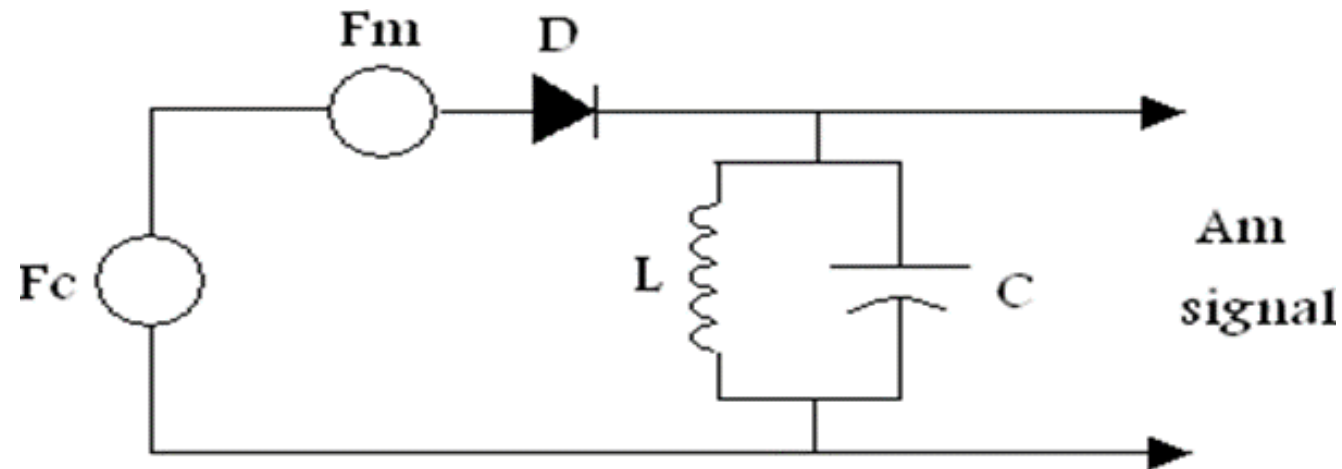
Audio amplifier: Used to amplify message signal.

Modulated Amplifier: This amplifier have non linear output in order to get all frequency component in the AM- signal.



Practical Circuit

The diode in the circuit represents the nonlinear element required to construct the AM signal.



Radio Receiver Essential Parameters of Amplitude Modulation

- Radio receivers are crucial devices in communication systems, enabling the reception of signals transmitted over long distances.
- In amplitude modulation (AM) systems, the quality and efficiency of the receiver depend on several essential parameters, such as: **Sensitivity, Selectivity, Fidelity (Audio Quality), and Signal-to-Noise Ratio (SNR).**

Radio Receiver Essential Parameters of Amplitude Modulation

1. Sensitivity

- Sensitivity refers to the ability of a radio receiver to detect weak signals.
- It is typically measured in microvolts (μV) or decibels-microvolts ($\text{dB}\mu\text{V}$).
- Higher sensitivity allows reception of distant or weak signals.
- It is defined as the minimum signal strength required to produce a specified signal-to-noise ratio (SNR) at the output.

2. Selectivity

- Selectivity is the ability of a receiver to differentiate between the desired signal and adjacent interfering signals.
- It is measured in decibels (dB) at specific frequency offsets from the carrier.
- High selectivity prevents interference from nearby stations.

3. Fidelity (Audio Quality)

- Fidelity determines how accurately the receiver reproduces the original audio signal.
- High fidelity ensures minimal distortion and high-quality sound output.
- Affected by the bandwidth of the **IF** and audio stages.
- Too narrow a bandwidth degrades sound quality, while too wide a bandwidth allows more noise and interference.

4. Signal-to-Noise Ratio (SNR)

- SNR is the ratio of the desired signal power to the noise power in the system.
- Measured in decibels (dB).
- A higher SNR means clearer reception and better quality audio.
- Improved by minimizing internal receiver noise and using high-quality components.

Generation of Double Sideband-Suppressed Carrier (DSB-SC) Modulation

- DSB-SC modulation eliminates the carrier signal, transmitting only the sidebands, thereby improving power efficiency.
- It can be generated using balanced modulators, such as: Cowan Modulator and Ring Modulator

Cowan Modulator

The Cowan modulator is a practical circuit that generates DSB-SC signals using a nonlinear device (such as a diode or transistor) combined with filtering techniques.

It works as follows:

1. The input signal (message) is applied to a modulator circuit containing a nonlinear device.
2. A local oscillator (LO) generates a high-frequency carrier wave.
3. The nonlinear device mixes the carrier with the message signal, producing sum and difference frequencies.
4. A band-pass filter removes the unwanted higher harmonics and isolates the DSB-SC signal.

RING MODULATOR

- A ring modulator is another method of generating DSB-SC signals using four diodes arranged in a ring configuration. Its operation includes:
 1. A balanced configuration ensures that the carrier is canceled out, leaving only the sidebands.
 2. The carrier and message signals are applied to a diode bridge circuit.
 3. The diodes operate as nonlinear switches, generating sum and difference frequency components.
 4. Filtering extracts the desired DSB-SC signal.
- Both Cowan and Ring modulators are widely used in AM systems where carrier suppression is necessary for efficient transmission.

Single Sideband (SSB) Modulation

- SSB modulation further improves bandwidth efficiency by eliminating one of the sidebands in a DSB-SC signal. This can be achieved by:
 1. **Filtering Method**: A high-quality band-pass filter removes the unwanted sideband after generating a DSB-SC signal.
 2. **Phase Shift Method**: Utilizes two balanced modulators and a phase shift network to cancel out one sideband.
- SSB modulation is crucial in applications like radio communications and voice transmission due to its reduced bandwidth requirements.

Demodulation of AM Signal

Demodulation is the process of extracting the original message signal from a modulated carrier.

1. Envelope Detector

An envelope detector is a simple and widely used method to demodulate standard AM signals (with a carrier). It operates as follows:

Rectification: A diode rectifies the incoming AM signal, allowing only positive half-cycles.

Filtering: A capacitor smooths the rectified signal, following its envelope.

Resistor Discharge: A resistor helps discharge the capacitor between signal peaks, preserving the original message waveform.

The output is the demodulated message signal.

Advantages

- Simple and low-cost.
- Effective for standard AM signals with a strong carrier.

Disadvantages

- Ineffective for DSB-SC and SSB signals.
- Distortion occurs if modulation depth is too high.

2. Synchronous Detector (Coherent Detection)

Synchronous detection is a more sophisticated method used for demodulating DSB-SC and SSB signals. It requires a locally generated carrier synchronized with the original transmitter carrier. The process includes:

Multiplication: The received AM signal is multiplied by a locally generated carrier.

Filtering: A low-pass filter extracts the original message signal by removing high-frequency components.

Advantages

Can demodulate DSB-SC and SSB signals.

Provides high-fidelity signal reproduction.

Disadvantages

Requires precise synchronization with the original carrier.

More complex than envelope detection.

NEXT LECTURE

Next lecture we will study the following:

1. Frequency Modulation
2. Frequency Modulation receiver
3. Frequency Modulation Generation

THE END

Middle Technical University

Electrical Engineering Technical College

Cyber Security Technology Engineering Department



COMMUNICATION FUNDAMENTALS

Second Stage

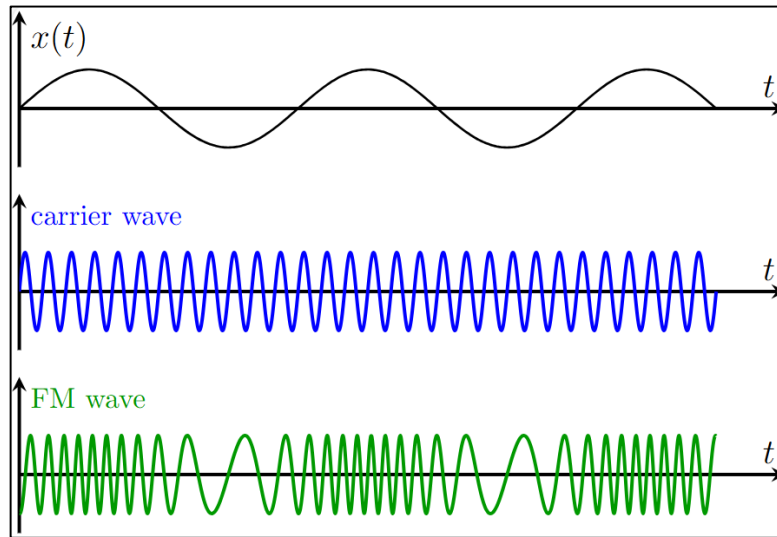
Frequency Modulation

Prepared by: Assist. Lec. Mustafa Ahmed Saadi

2025 – 2024

FREQUENCY MODULATION

Frequency Modulation is a method of encoding information on one carrier wave by changing the wave carrier frequency. In frequency modulation, the frequency of the carrier wave is changed according to the frequency of the modulating signal. Where, the carrier frequency **increases** when the message signal is positive and **decreases** when the message signal is negative.



The amount by which the carrier frequency varies from its resting frequency due to modulation is called **Frequency Deviation (Δf)**. It is directly proportional to the amplitude of the modulating signal and determines the overall bandwidth of the FM signal. The peak frequency deviation is given by:

$$\Delta f = K_f V_m$$

Where:

- K_f is the frequency sensitivity of the modulator (Hz/V).
- V_m is the peak amplitude of the modulating signal.

FREQUENCY MODULATION INDEX

The **frequency modulation index** (denoted by β) is a key parameter in **Frequency Modulation (FM)**, which indicates the extent of frequency deviation relative to the frequency of the modulating signal. It determines how much the carrier frequency changes in response to the modulating signal. The modulation index β is defined as:

$$\beta = \frac{\Delta f}{f_m}$$

Where:

- β = Modulation index (unitless)
- Δf = Frequency deviation (Hz), the maximum shift in frequency from the carrier frequency due to modulation
- f_m = Frequency of the modulating signal (Hz)

Key Points:

1. **If $\beta = 1$:** The frequency deviation is equal to the frequency of the modulating signal.
2. **If $\beta > 1$:** The frequency deviation is greater than the modulating signal frequency, resulting in a wider frequency spectrum.
3. **If $\beta < 1$:** The frequency deviation is smaller than the modulating signal frequency, leading to a narrower frequency spectrum.

The modulation index affects the **bandwidth** and **spectral characteristics** of the FM signal:

- Larger β results in a broader bandwidth and more sidebands.
- Smaller β results in a narrower bandwidth.

Thus, the modulation index plays a significant role in determining the quality, bandwidth, and efficiency of FM systems.

MATHEMATICAL REPRESENTATION OF FREQUENCY MODULATION

In frequency modulation, the instantaneous frequency of the carrier signal is varied in proportion to the message signal. Let's define the key components mathematically:

1. Carrier Wave

A high-frequency sinusoidal signal:

$$c(t) = V_c \cos(2\pi f_c t)$$

where:

V_c = Amplitude of the carrier wave

f_c = Carrier frequency

t = Time

2. Modulating (Message) Signal

The information signal that contains the desired message:

$$m(t) = V_m \cos(2\pi f_m t)$$

where:

V_m = Amplitude of the message wave

f_m = message frequency

3. Frequency Modulated (FM) Signal

The FM signal is formed by varying the carrier frequency according to the message signal:

$$V_{FM}(t) = V_c \cos(2\pi f_c t + \beta \sin(2\pi f_m t))$$

DEVIATION RATIO

The **Deviation Ratio** is a key parameter in frequency modulation (FM) that measures the relationship between the frequency deviation of the carrier signal and the frequency of the

modulating signal. It is used to quantify the extent to which the carrier's frequency deviates due to modulation. The **Deviation Ratio** is given by:

$$\text{Deviation Ratio} = \frac{\Delta f}{f_m}$$

where:

- Δf = Frequency deviation (Hz), the maximum shift in the carrier frequency caused by modulation
- f_m = Frequency of the modulating signal (Hz)

Example 1: If the frequency deviation Δf is 75 kHz and the modulating signal frequency f_m is 15 kHz, find the deviation ratio.

$$\text{Deviation Ratio} = 75 \text{ kHz} / 15 \text{ kHz} = 5$$

This means the carrier frequency deviates by 5 times the frequency of the modulating signal.

CARRIER SWING

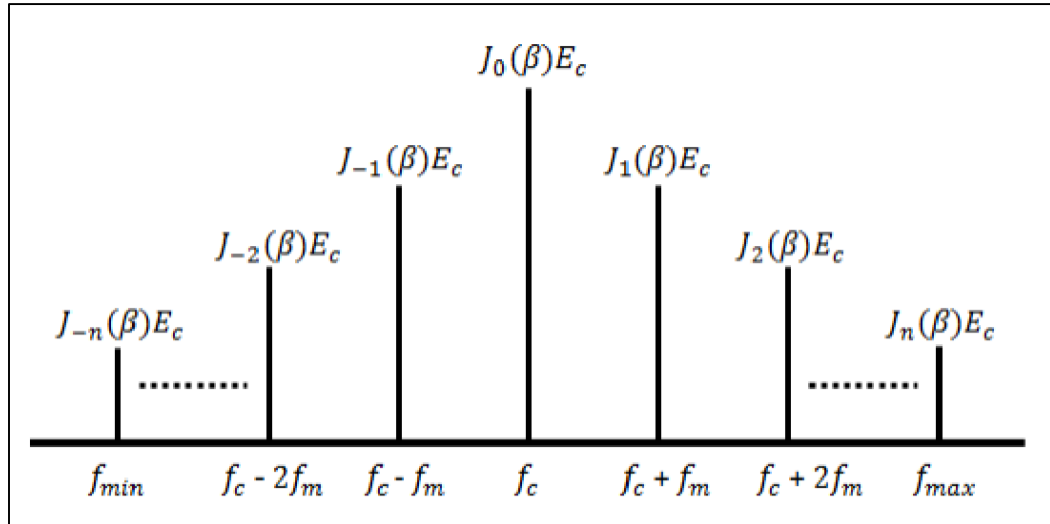
Carrier swing in frequency modulation refers to the total frequency range over which the carrier frequency oscillates due to the modulation. It represents the difference between the maximum and minimum frequencies of the modulated signal.

Mathematically, the **carrier swing** is given by:

$$\text{Carrier Swing} = 2 \times \Delta f$$

Frequency spectrum of FM

Any signal that is modulated produces sidebands. In the case of an amplitude modulated signal they are easy to determine, but for frequency modulation is not quite easy. They are dependent upon not only the deviation, but also the level of deviation, i.e. the modulation index β . The total spectrum is an infinite series of discrete spectral components expressed by a complex formula using Bessel functions of the first kind.



BANDWIDTH

In frequency modulation (FM), **bandwidth** refers to the range of frequencies that the modulated signal occupies. FM is a type of modulation where the frequency of the carrier wave is varied in proportion to the instantaneous amplitude of the input signal (the message signal).

The bandwidth of an FM signal depends on **Modulation Index (β)**, where:

1. $\beta > 1$: narrow band FM

$$BW_{FM} = 2 \times f_m$$

2. $\beta < 20$: wide band FM

$$BW_{FM} = 2 \times \Delta f$$

3. $1 < \beta < 20$: normal band FM

$$BW_{FM} \approx 2 \times (\Delta f + f_m)$$

Where:

- BW_{FM} is the bandwidth,
- Δf is the peak frequency deviation,

- f_m is the highest frequency of the modulating signal.

In FM, the bandwidth is often wider than in amplitude modulation (AM), especially for high modulation indices. This broader bandwidth is one reason why FM is more resistant to noise and interference compared to AM.

Important equations:

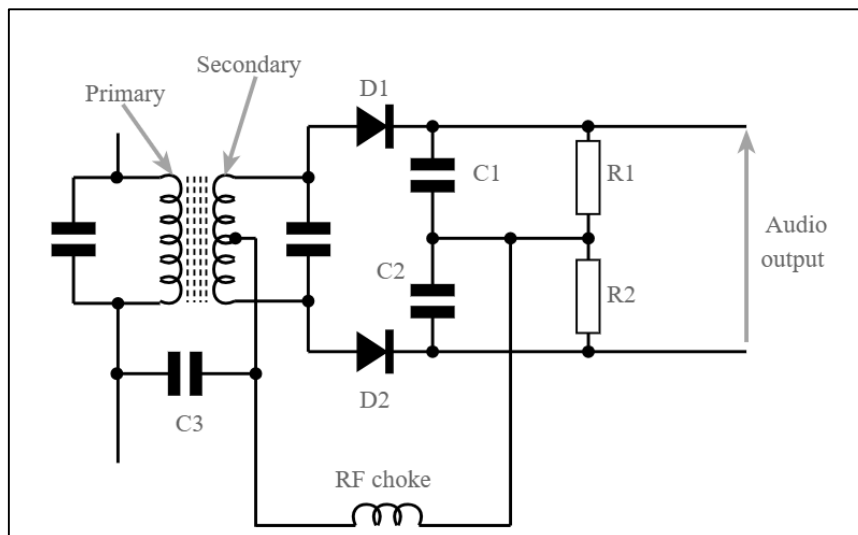
- First Lower sideband (minimum frequency) $\rightarrow f_{min} = f_c - f_m$
- First Upper sideband (maximum frequency) $\rightarrow f_{max} = f_c + f_m$
- Carrier frequency $\rightarrow f_c = \frac{f_{max} + f_{min}}{2}$
- Carrier swing $= f_{max} - f_{min}$

FM Receiver: FM Discriminator

To demodulate an FM signal, a device known as an **FM discriminator** or **detector** is used. The two primary types of FM discriminators are:

1. Foster-Seeley Discriminator

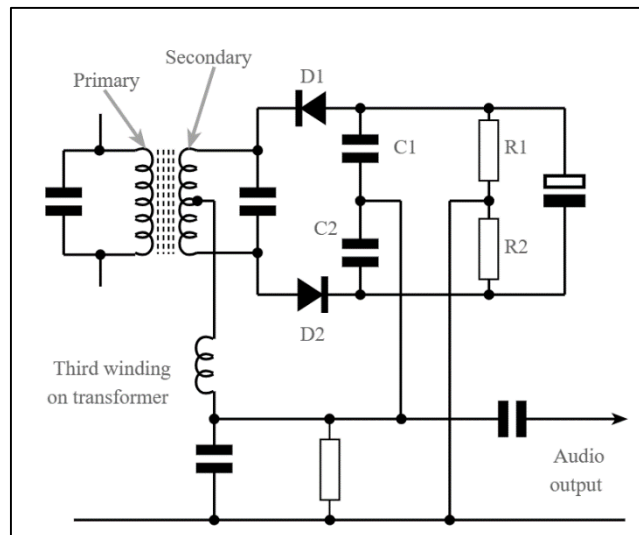
The **Foster-Seeley discriminator** is a frequency-to-voltage converter that operates using a double-tuned RF transformer and a diode detector. It provides an output voltage proportional to the frequency deviation of the FM signal. This circuit is widely used in radio receivers due to its simplicity and good linearity.



2. Ratio Detector

The **Ratio Detector** is another type of FM discriminator that uses a similar double-tuned transformer but differs in its diode arrangement. It provides better noise immunity than the Foster-

Seeley discriminator and does not require a limiter stage, making it advantageous for FM broadcast receivers.



FM Generation

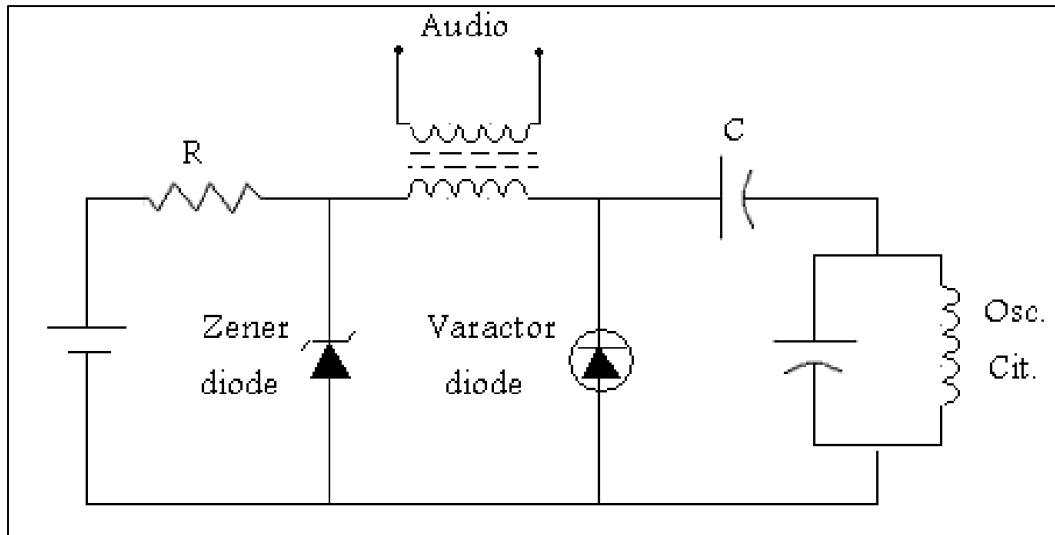
There are two types for circuits, used to generate FM signal: Direct method and indirect method

1. Direct Method:

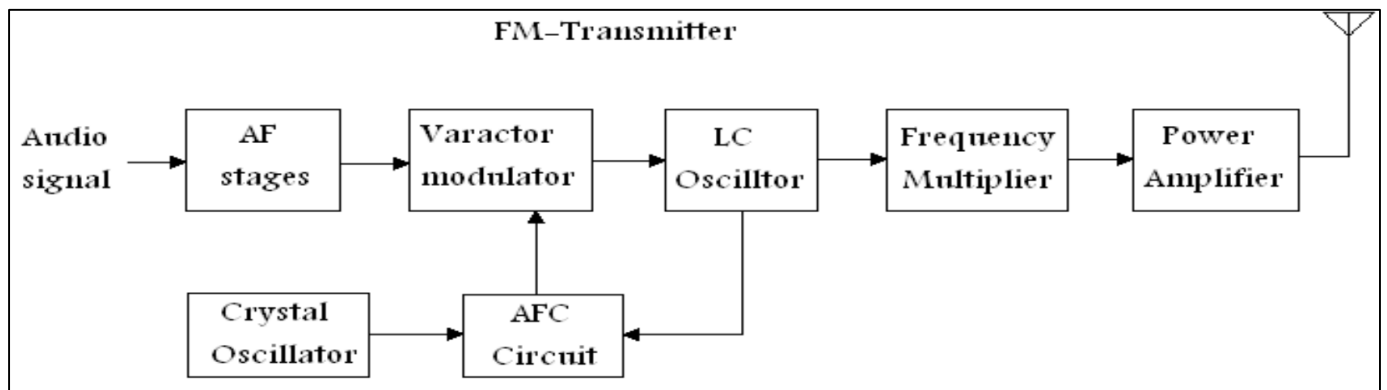
In this method the carrier frequency is directly varied in accordance with the incoming message signal. In which the carrier frequency is varying by means of modulating signal, this can be done by using non-linear device such as a valve or transistor to vary the capacitance across the tuned circuit of an oscillator.

1.1. Varactor modulator

We use the variable capacitance of a P-N junction semiconductor which is known as a (varactor) diode. When a d.c. voltage is applied in reverse direction across the p-n junction, a depletion layer is developed, hence, a capacitance appears across the junction between the separated charges and may be varied by varying the applied d.c. voltage. Where $C \propto 1/V$. Selection diode has $C=150\text{-}200\text{ pF}$ with 1 V and C is varying $(10\text{-}15)\text{ pF/volt}$. A typical varactor modulator is shown in figure below.



The audio signal is amplified and used to derive a varactor modulator which is used to change the frequency of LC oscillator in which its center frequency is stabilize by an AFC loop, the output frequency is multiplied several times to reach VHF band, then the FM signal is amplified using R.F. class C power amplifier, then radiated by antenna.



Example: An oscillator operates at 100 MHz has 75 pF in its T.C. what total capacitance swing (ΔC) must the varactor supply to produce $\Delta f = 80$ KHz.

Solution:

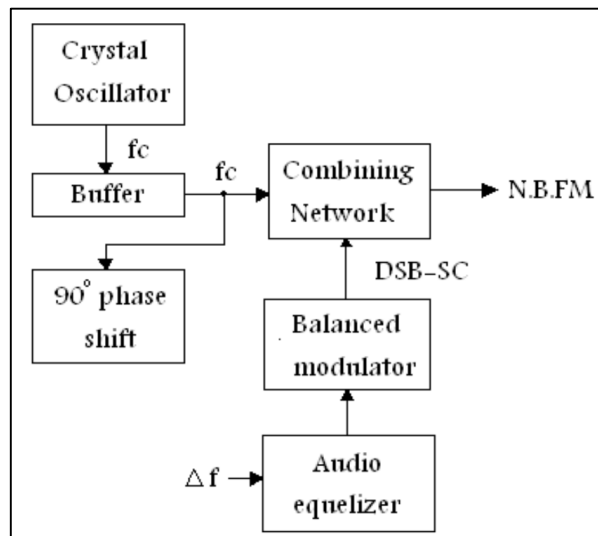
$$\Delta C = \frac{2 \times \Delta f \times C}{\Delta f}$$

$$\Delta C = \frac{2 \times 80 \times 10^3 \times 75 \times 10^{-12}}{100 \times 10^6}$$

$$\Delta C = 0.12 \text{ PF}$$

2. Indirect Method :

This method was first proposed by Armstrong. In this method the modulating wave is first used to produce a narrow-band FM wave, and frequency multiplication is next used to increase the frequency deviation to the desired level. To overcome the problem of oscillator stability in the FM generation, of the direct method, we use to generate FM indirectly, using phase modulation (PM) (Armstrong system)



As in above figure the diagram gives narrow band FM at the output of the combining network, we could use further stages to expand the FM signal BW. The basic feature of Armstrong system is to phase modulate a carrier which has been shifted 90° relative to the modulating component, the phase shift is small (< 50 rad) to reduce distortion this make $\Delta f = 75$ KHz by several stages of multipliers.

Example 2:

A 107.6 MHz carrier is modulated by 7 KHz sin wave, the FM signal produced has frequency deviation of 50 KHz.

1. find the carrier Swing of the FM signal
2. What are the heights and the lowest modulated signal frequencies?
3. Modulation index.

Sol

$\therefore F_c = 107.6 \text{ MHz}, f_m = 7 \text{ KHz}, \Delta f = 50 \text{ KHz}$

1. $S.C = 2 \Delta f = 2 \times 50 = 100 \text{ KHz}$
2. $f_H = f_c + \Delta f = 107.6 \times 10^6 + 50 \times 10^3 = 107650 \text{ KHz}$
 $f_L = f_c - \Delta f = 107.6 \times 10^6 - 50 \times 10^3 = 107550 \text{ KHz}$
3. $m_f = \frac{\Delta f}{f_m} = \frac{50 \times 10^3}{7 \times 10^3} = 7.143$

Example 3:

10 MHz carrier frequency is modulated by a sine wave, to give frequency deviation of 50 KHz, determine the approximate BW, if the modulating frequency is: 500 KHz, 500 Hz, 10 KHz

Sol

1. $\therefore m_f = \frac{\Delta f}{f_m} = \frac{50 \times 10^3}{500 \times 10^3} = 0.1$
 $\therefore BW = 2 f_m = 2 \times 500 \times 10^3 = 1 \text{ MHz}$
2. $m_f = \frac{50 \times 10^3}{500} = 100$
 $BW = 2 \Delta f = 2 \times 50 = 100 \text{ KHz}$
3. $m_f = \frac{50}{10} = 5$
 $BW = 2(\Delta f + f_m) = 2(50 \times 10^3 + 10 \times 10^3) = 120 \text{ KHz}$

Example 4:

FM signal is modulated by 3 KHz sine wave reaches maximum frequency of 100.02 MHz and minimum frequency of 99.98 MHz, find: C.S, f_c , Δf , m_f .

Sol

1. $C.S = f_{\max} - f_{\min} = 100.02 - 99.98 = 0.04 \text{ MHz} = 40 \text{ KHz}$.
2. carrier frequency is the midway between max & min frequency reached by FM wave

$$f_c = \frac{f_{\max} + f_{\min}}{2} = \frac{100.02 + 99.98}{2} = 100 \text{ MHz}$$

$$3. \quad \Delta f = \frac{C.S}{2} = \frac{40 \times 10^3}{2} = 20 \text{ KHz}$$

$$4. \quad m_f = \frac{\Delta f}{f_m} = \frac{20}{3} = 6.667 \text{ KHz}$$

Example 5:

Determine Δf , C.S, f_L if $f_c = 105 \text{ MHz}$ and $f_H = 105.007 \text{ MHz}$

Sol

1. $f_H = f_c + \Delta f \Rightarrow \Delta f = f_H - f_c = 105.007 - 105 = 7 \text{ KHz}$
2. $C.S = 2 \times \Delta f = 2 \times 7 = 14 \text{ KHz}$
3. $f_L = f_c - \Delta f = 105 \times 10^3 - 7 = 104993 \text{ KHz}$

Example 6:

Find the modulation index if C.S = 100 KHz and $f_m = 8 \text{ KHz}$

Sol

$$\therefore C.S = 2 \Delta f \Rightarrow \Delta f = \frac{C.S}{2} = \frac{100}{2} = 50 \text{ KHz}$$

$$\therefore m_f = \frac{\Delta f}{f_m} = \frac{50 \text{ KHz}}{8 \text{ KHz}} = 6.25$$

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COMMUNICATION FUNDAMENTALS

Second Stage

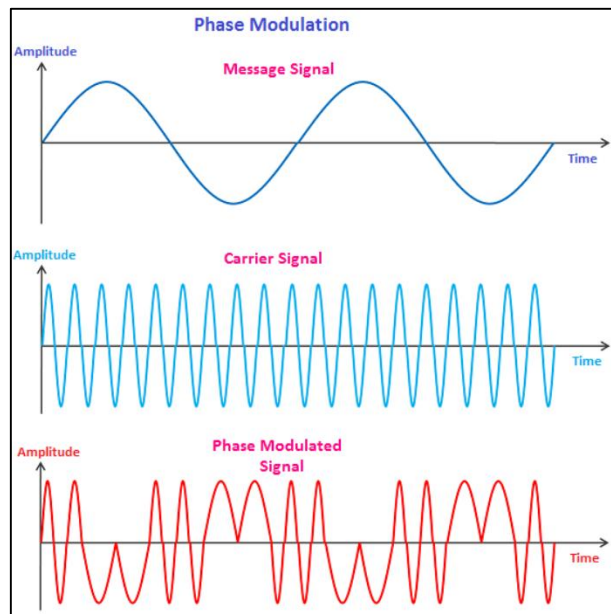
Phase Modulation

Prepared by: Assist. Lec. Mustafa Ahmed Saadi

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INTRODUCTION

Phase Modulation (PM) is a type of modulation where the phase of the carrier signal is varied in proportion to the instantaneous value of the message (modulating) signal. Phase Modulation is used in various applications such as Digital Modulation Schemes, Telemetry and Satellite Communication, and Noise-resilient communication systems. The figure below shows phase modulation:



SIGNAL EQUATIONS

Message signal:

$$m(t) = V_m \cos(2\pi f_m t)$$

Carrier Signal:

$$c(t) = V_c \cos(2\pi f_c t)$$

PM Signal:

$$s(t) = V_c \cos(2\pi f_c t + K_p \cdot m(t))$$

where:

K_p : phase sensitivity

BANDWIDTH and MODULATION INDEX

The bandwidth of PM depends on the amplitude and frequency of the message and the phase sensitivity k_p . The bandwidth can be calculated using Carson's rule:

$$\text{BW} \approx 2 (\Delta f + f_m)$$

Where:

Δf = peak frequency deviation

f_m = highest frequency in the message signal

The equation of frequency deviation is:

$$\Delta f = K_p V_m f_m$$

The modulation index (μ) on phase modulation is:

$$\mu = K_p V_m$$

The same equation is phase deviation $\Delta\phi$:

$$\Delta\phi = \mu = K_p V_m$$

Example: In phase modulation, if the message $m(t) = 3 \cos(600\pi t)$ signal is modulated using the carrier $c(t) = 6 \cos(30000\pi t)$ and the phase sensitivity is 0.5 then find:

1. Modulation index
2. Modulated signal
3. Bandwidth

Solution:

1.

$$\mu = 0.5 * 3$$

$$\mu = 1.5$$

2.

$$s(t) = V_c \cos(2\pi f_c t + K_p \cdot m(t))$$
$$s(t) = 6 \cos(30000\pi t + 1.5 \cos(600\pi t))$$

3.

$$BW \approx 2 (\Delta f + f_m)$$

$$\Delta f = K_p V_m f_m$$

$$f_m = 600\pi / 2\pi = 300 \text{ Hz}$$

$$\Delta f = 0.5 * 3 * 300$$

$$\Delta f = 450$$

$$BW \approx 2 (450 + 300)$$

$$BW \approx 1500 \text{ Hz}$$

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COMMUNICATION FUNDAMENTALS

Second Stage

Introduction to Digital modulation and digital coding

Prepared by: Assist. Lec. Mustafa Ahmed Saadi

2025 – 2024

INTRODUCTION

Digital communication systems have transformed the way information is transmitted and received, offering significant advantages over traditional analog systems. Two key concepts that form the foundation of digital communications are **digital modulation** and **digital coding**. Digital modulation involves converting digital data into analog signals for transmission, while digital coding ensures that the data is efficiently represented and protected against errors. Together, these techniques enable reliable and high-speed communication across various media, including wireless networks, fiber optics, and satellite links. Common digital modulation techniques include:

1. **Amplitude Shift Keying (ASK)**
2. **Frequency Shift Keying (FSK)**
3. **Phase Shift Keying (PSK)**
4. **Quadrature Amplitude Modulation (QAM).**

Each method has its trade-offs in terms of complexity, bandwidth efficiency, and resistance to interference. For example, while ASK is simple to implement, it is highly sensitive to noise, whereas QAM offers higher data rates but requires more sophisticated demodulation techniques. The three primary digital modulation are Amplitude Shift Keying (ASK), Phase Shift Keying (PSK), and Frequency Shift Keying (FSK). These methods used in modern communication systems including Wi-Fi, satellite communication, radio systems, and mobile networks. Where digital modulation is the process of encoding digital information onto an analog carrier signal. The analog carrier can be manipulated in amplitude, frequency, or phase to represent binary data (0s and 1s).

1. Amplitude Shift Keying (ASK)

ASK is a form of amplitude modulation that represents binary data as variations in the amplitude of a carrier wave.

How it works:

- Binary 1 is represented by transmitting a carrier wave of a specific amplitude.
- Binary 0 is represented by either no carrier (on-off keying) or a lower amplitude signal.

Mathematical Representation:

$$s(t) = d(t) \cdot V \cos(2\pi f_c t)$$

where:

- V is the amplitude of the carrier
- f_c is the carrier frequency
- $d(t)$ is the digital data, $d(t) \in \{0,1\}$

Advantages:

- Simple implementation.
- Efficient in terms of bandwidth.

Disadvantages:

- Highly susceptible to noise and interference.
- Poor performance in noisy channels.

Applications:

- Early modems
- Optical fiber communication (On-Off Keying).

2. Phase Shift Keying (PSK)

PSK is a digital modulation scheme that conveys data by changing the phase of the carrier signal.

How it works:

Each symbol represents a fixed number of bits using a distinct phase of the carrier.

$$s(t) = V \cos(2\pi f_c t + \pi d(t))$$

Advantages:

- More robust against noise compared to ASK.
- Efficient bandwidth usage.

Disadvantages:

- Requires coherent detection (carrier synchronization).
- More complex hardware implementation.

Applications:

- Satellite communication.
- Wi-Fi (IEEE 802.11b).
- GSM cellular systems.

3. Frequency Shift Keying (FSK)**Definition:**

FSK is a digital modulation technique where digital information is transmitted through discrete frequency changes of a carrier wave.

How it works:

- Binary 0 and 1 are represented by two distinct frequencies.
- For higher-order FSK (e.g., 4-FSK), more frequencies are used to encode more bits per symbol.

Mathematical Representation:

$$s(t) = V \cos (2\pi f_i t)$$

- f_i = frequency assigned to a particular bit or group of bits.

Advantages:

- Resistant to signal-level noise.
- Good performance in wireless environments.

Disadvantages:

- Requires a wider bandwidth than ASK and PSK.
- More complex receiver design (especially for coherent FSK).

Applications:

- Radio communication (e.g., walkie-talkies).
- Caller ID systems.
- Low-power devices and remote sensors.

